Model specification in tempssm

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Overview

This vignette describes the linear Gaussian state-space model implemented in `tempssm`. The package is designed for temperature time series and represents observed variation as a combination of long-term trend, seasonal variation, autoregressive dependence, and optional exogenous effects.

The model is estimated with the `KFAS` package as the computational backend. The main fitting function, `tempssm()`, returns both filtering and smoothing estimates. Unless otherwise stated, summaries, diagnostics, and plots shown by the package are based on smoothed state estimates.

State-Space Model

We consider an extended version of the Basic Structural Time Series Model (BSTSM) to describe temperature time series with explicit long-term trend, seasonal variability, and autoregressive structure. The BSTSM is a standard state-space formulation that represents an observed time series using latent trend, seasonal, and irregular components (Harvey 1989). Following the temperature time-series application of Baba et al. (2024), the model used in `tempssm` keeps this interpretable decomposition and adds autoregressive dependence and optional exogenous effects. This structure is useful for separating long-term temperature change, seasonal cycles, and short-term departures.

The observation equation is given by

$$y_t = \alpha_t + v_t. \tag{1}$$

Here, t denotes the time index, y_t is the observed temperature, α_t is the latent state, and v_t is the observation error.

The latent state is decomposed as

$$\alpha_t = \mu_t + s_t + r_t + E_t. \tag{2}$$

Here, μ_t is the long-term trend component, s_t is the seasonal component, r_t is a stationary autoregressive component, and E_t is the contribution of exogenous variables.

Long-Term Trend

The level component follows a second-order stochastic process:

$$\mu_t = 2\mu_{t-1} - \mu_{t-2} + \zeta_t, \qquad \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2). \tag{3}$$

Here, ζ_t is the process error of the long-term trend component. This formulation allows the long-term level and its rate of change to vary over time.

Seasonal Component

The seasonal component is modeled with a sum-to-zero constraint:

$$s_t = -\sum_{i=t-f}^{t-1} s_i + \omega_t, \qquad \omega_t \sim \mathcal{N}(0, \sigma_\omega^2). \tag{4}$$

Here, ω_t is the process error of the seasonal component and f denotes the seasonal frequency, for example $f = 12$ for monthly data. This formulation makes seasonal effects identifiable and enables models to be constructed for time series with different temporal resolutions.

By imposing the sum-to-zero constraint over one complete seasonal cycle, the seasonal component captures recurring deviations from the underlying long-term trend without introducing long-term drift. In models without a seasonal component, the seasonal term s_t is set to zero and omitted from the state equation.

Autoregressive Component

The autoregressive component follows an lth-order autoregressive (AR) process:

$$r_t = \phi_1 r_{t-1} + \phi_2 r_{t-2} + \cdots + \phi_l r_{t-l} + \tau_t, \qquad \tau_t \sim \mathcal{N}(0, \sigma_\tau^2). \tag{5}$$

Here, τ_t denotes the process error of the autoregressive component. The process error terms (ζ_t, ω_t, and τ_t) are assumed to be mutually independent and normally distributed.

The autoregressive component is mainly intended to absorb remaining short-term serial dependence after accounting for the long-term trend and seasonal structure. In applied analyses, low-order AR structures are often useful as starting points, and higher orders should be considered together with residual diagnostics and prediction performance.

Exogenous Component

The exogenous component is defined as

$$E_t = \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_m x_{m,t}. \tag{6}$$

The exogenous term E_t represents the influence of external factors. These may include large-scale climate indices, regional environmental variables, or other physically motivated predictors relevant to the observed temperature time series. Each exogenous variable enters the model linearly through a time-invariant regression coefficient $(\beta_1, \beta_2, \dots, \beta_m)$, allowing the magnitude and direction of its contribution to be estimated jointly with the latent state components.

For models with exogenous variables, prediction beyond the observed response period requires future exogenous values or an explicit assumption about those values.

Number of Estimated Parameters

When applying the model to observational data, the total number of parameters to be estimated (k) depends on the order of the autoregressive component and the number of exogenous variables included in the model.

For example, in a model with a second-order autoregressive component and no exogenous covariates, six parameters are estimated: the observation error variance, the process error variances associated with the long-term trend, seasonal, and autoregressive components, and the first- and second-order autoregressive coefficients (ϕ_1 and ϕ_2).

In the general case with an lth-order autoregressive component and m exogenous variables, the total number of parameters is

$$k = 4 + l + m.$$

Parameter Estimation

The core implementation of the parameter estimation procedure is based on the supplementary code provided by Baba et al. (2024). Parameter estimation is performed using a two-step optimization strategy recommended by Helske (2017).

In the first step, model parameters are estimated using the Nelder-Mead method (Nelder & Mead, 1965) with user-specified initial values. The resulting estimates are then used as initial values in a second optimization step based on the BFGS algorithm (Shanno, 1970).

By default, `tempssm()` uses the marginal likelihood implemented in KFAS for parameter estimation. The diffuse likelihood remains available by explicitly setting `marginal = FALSE`. The selected likelihood type is stored in the fitted `tempssm` object and is used consistently by downstream methods such as `logLik()` and `summary()`.

References

Baba, S., Ishii, H., and Yoshiyama, T. (2024). Estimating sea temperature trends using a linear Gaussian state-space model in Jogashima, Kanagawa, Japan. *Bulletin of the Japanese Society of Fisheries Oceanography*, 88(3), 190-199. (In Japanese with an English abstract.) https://doi.org/10.34423/jsfo.88.3_190

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